

ALI AJVAND

Data Scientist | ML Engineer | BI Developer

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[2342b3232 · github.com/aliajvand](https://github.com/aliajvand) · aliajvand.github.io/portfolio Available for full-time remote or on-site roles

PROFESSIONAL SUMMARY

Data Scientist with end-to-end experience in Python/SQL pipelines, machine learning, forecasting, and BI. Builds leakage-safe, chronologically validated models benchmarked against real operational baselines: delivered 9.36% WMAPE demand forecasts (vs 34.06% heuristic), won 2nd place in Hackathon Pel 4 (AI Innovation Factory × Jobama) with 82% prediction accuracy on a real-world hospitality demand problem, and shipped a calibrated credit scorecard with KS 0.427 and 9.8% approved-book bad rate. Strong in probability calibration, SHAP explainability, cost-sensitive decision policies, LLM-based tooling (LangChain, PyTorch, Hugging Face) and MLOps (Docker, CI/CD, pytest).

TECHNICAL SKILLS

Languages & Data: Python, SQL, T-SQL, DAX, pandas, MySQL, PostgreSQL, SQL Server, DuckDB, Parquet, ETL/ELT

Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, forecasting, classification, regression, clustering, SHAP, feature engineering, probability calibration (isotonic, Platt)

Deep Learning & NLP: PyTorch, Sentence-Transformers, Hugging Face Transformers, FAISS, LangChain (LLM orchestration & retrieval pipelines)

Statistics & Validation: walk-forward & grouped cross-validation, leakage auditing, A/B testing, hypothesis testing, cost-sensitive evaluation

BI & Visualization: Power BI (star schema, DAX, drill-through, KPI dictionaries), Tableau, Metabase, Matplotlib, Seaborn, Plotly, Folium

Engineering & MLOps: FastAPI, Flask, Streamlit, Docker, Git, GitHub Actions CI/CD, pytest, ruff, mypy, reproducible pipelines

KEY ACHIEVEMENTS

- Demand forecasting: cut retail demand forecast error from 34.06% to 9.36% WMAPE at a 42-day horizon across 1,115 stores, holding bias to -1.9% and bringing 98.3% of the estate inside the 15% automation threshold.
- Hackathon Pol 4 (2nd place): won 2nd place in a 72-hour team hackathon (AI Innovation Factory × Jobama) on hospitality demand forecasting — built a time-series ensemble model reaching 82% prediction accuracy on real booking data from 321 Iranian cities, delivering actionable early-warning signals for capacity and pricing decisions.
- Credit risk modeling: ROC-AUC 0.780, KS 0.427, Brier 0.135 calibrated scorecard on 30,000 clients; riskiest decile defaults at 70.0% versus a 22.1% portfolio rate (3.17x lift).
- BI & automation: Power BI dashboards and automated Python reporting workflows that cut manual reporting effort by 40% for field sales managers.
- LLM & RAG pipeline: designed and deployed a LangChain-based retrieval-augmented generation (RAG) pipeline for an automotive-sector startup, integrating FAISS vector search and Hugging Face embeddings to retrieve structured vehicle data and generate accurate, context-aware responses.

PROFESSIONAL EXPERIENCE

Data Analyst (Project-Based) — BehinBazar, Tehran, Iran

Apr 2024 – Nov 2024

- Built end-to-end ETL pipelines and Power BI dashboards for JTI field-sales operations, giving management a same-day view of sales performance, coverage, and team productivity.
- Automated weekly reporting workflows with Python and Google Sheets API, cutting manual reporting effort by 40% and eliminating copy-paste errors.
- Developed Folium geospatial tools for store-coverage analysis and route planning, converting manual scheduling into a repeatable, data-driven process.
- Partnered with field teams to consolidate scattered progress updates into a single actionable report, improving data quality across all downstream analytics.

SELECTED DATA SCIENCE & MACHINE LEARNING PROJECTS

Demand & Sales Forecasting — Hierarchical Retail Time-Series Pipeline

github.com/aliajvand/demand_sales_forecasting

- Forecasted 844,338 open store-days across 1,115 stores, validated on three non-overlapping 42-day walkforward folds where training data always ends before the fold begins.
- Beat the planner's same-weekday heuristic by 24.7 points (34.06% → 9.36% WMAPE), holding bias to -1.9% against a ±5% target.
- Published per-store error (median 8.7%, p90 11.9%) to support differentiated safety-stock policies rather than a single chain-wide average.

Hackathon Pel 4 — Demand Forecasting (2nd Place)

AI Innovation Factory × Jobama

- Competed in a 72-hour team hackathon focused on real-world hospitality demand prediction for Iranian hotel booking platforms.
- Built a time-series ensemble model (LightGBM, XGBoost, CatBoost, HistGradientBoosting) integrating accrual-based nowcasting, calendar seasonality, and geographic features across 321 cities.
- Achieved 82% prediction accuracy and designed early-warning capabilities to signal when to increase inventory and pricing 2–4 weeks in advance of demand peaks.
- Placed 2nd in the competition, demonstrating rapid prototyping, cross-functional teamwork, and delivery under pressure.

Credit Risk Decision Engine — Calibrated PD Scorecard & Approval Policy

github.com/aliajvand/credit_risk_decision_engine

- Built a calibrated default-risk workflow for 30,000 credit-card clients using 41 behavioral features: ROC-AUC 0.780, PR-AUC 0.554, KS 0.427, Brier 0.135, versus 0.703 ROC-AUC for a single-rule baseline.
- Optimized a 5:1 false-approval-to-false-decline cost policy, approving 54.4% of applicants at a 9.8% bad rate versus 15.9% under a naive PD < 0.50 rule; the riskiest decile defaults at 70.0% versus 22.1% portfolio-wide.
- Excluded demographic attributes (sex, education, marital status, age) from the feature matrix, retaining them only for adverse-impact monitoring, and published a re-costable threshold table.

Olist Delivery Performance Analytics — Predictive Logistics & Decision Policies

github.com/aliajvand/olist-delivery-analytics

- Built an end-to-end pipeline over ~99,000 Brazilian e-commerce orders (3,000 sellers, 27 states) to predict late delivery, estimate freight cost, and set promise dates using only purchase-time information.
- Engineered leakage-safe seller, route, and customer history features via as-of joins keyed to purchase time, verified through a 39-feature leakage audit and 34 automated tests.
- Translated five models into operational policies — capacity-constrained expedite queues and quantile promise dates — with isotonic calibration, cost-sensitivity tables, and incumbent baselines.

EV & Automotive Analytics — Market Intelligence & Price Regression

github.com/aliajvand/ev_automotive_analytics

- Analyzed 289,555 EV registrations in a Power BI star schema: 80.7% BEV share and 69.3% fleet concentration across the top three counties, informing charging-infrastructure prioritization.
- Ran a leakage audit showing an apparent ROC-AUC of 1.00 fell to 0.66 after removing drivetrain-derived proxy features, preventing a misleading model claim from shipping.
- Priced 15,409 used-car listings on a nameplate-grouped split: explainable linear regression achieved R^2 0.83 and 19.3% MAPE versus 41.8% for a brand-median baseline and 20.8% for gradient boosting.

ADDITIONAL ANALYTICS WORK

Power BI e-commerce sales dashboard on MySQL · K-means segmentation · FP-Growth market-basket analysis · A/B testing & ad-spend analysis · RAG clinical decision-support API (FastAPI, FAISS, Sentence-Transformers, PyTorch) · LangChain-based LLM orchestration pipeline for a confidential automotive-sector startup project (in progress). Full case studies: aliajvand.github.io/portfolio

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Engineering — Islamic Azad University, Science and Research Branch, Tehran

Expected 2026

GPA: 16/20. Relevant coursework: data structures and algorithms, databases, statistics and probability, linear algebra, software engineering.

Certifications: Data Scientist Bootcamp — Daneshkar, 100+ hours (2025) · Business Intelligence Bootcamp — Udemy (2023) · Data Analyst Bootcamp — Alex The Analyst (2023) · LangChain for LLM Application Development — freeCodeCamp (2026)

HONORS & AWARDS

2nd Place — **Hackathon Pol 4** (AI Innovation Factory × Jobama) | Demand Forecasting | 72-hour Team Competition | 2026

LANGUAGES

Persian: Native · English: Professional working proficiency